

TPROFEDO8 – Assessment and Evaluation of Student Learning

Writing of Test Items Based on the Table of Specifications

Instruction:

Construct a 30-item test based on the TOS that you prepared. Use a multiple-choice item format.



Procedure

COMPETENCIES/ CONTENT	LEARNING OBJECTIVES	COGNITIVE LEVEL	ITEM PLACEMENT	NAME OF STUDENT WHO WROTE THE ITEM	ITEM	ANSWER KEY	JUSTIFICATION FOR THE CORRECT ANSWER
Components of an ecosystem	Identify which of the things found in the environment are biotic or abiotic	Remembering	1	Rose Marianne Louis M. Gardoce	_____ 1. Which of the following is a biotic component of an ecosystem? A. Water B. Sunlight C. Grass D. Soil	C. Grass	Grass is a living organism, so it is a biotic component of an ecosystem. Water, sunlight, and soil are nonliving or abiotic components.
Components of an ecosystem	Identify which of the things found in the environment are biotic or abiotic	Remembering	2	Rose Marianne Louis M. Gardoce	_____ 2. Which of the following is an abiotic component of an ecosystem? A. Bird B. Tree C. Mushroom D. Temperature	D. Temperature	Temperature is a nonliving environmental factor , making it an abiotic component of an ecosystem.

Components of an ecosystem	Identify which of the things found in the environment are biotic or abiotic	Remembering	3	Rose Marianne Louis M. Gardoce	_____ 3. Which of the following is an example of a living component of an ecosystem? A. Rock B. Air C. Bacteria D. Water	C. Bacteria	Bacteria are living organisms. Therefore, they are classified as biotic components of an ecosystem.
Components of an ecosystem	Identify which of the things found in the environment are biotic or abiotic	Remembering	4	Rose Marianne Louis M. Gardoce	_____ 4. Which of the following is NOT an abiotic factor? A. Sunlight B. Soil C. Water D. Grass	D. Grass	Grass is a living organism and is therefore biotic , not abiotic. Sunlight, soil, and water are abiotic factors.
Trophic levels	Identify which of the things found in the environment are biotic or abiotic	Remembering	5	Rose Marianne Louis M. Gardoce	_____ 5. Which organism occupies the first trophic level in a food chain? A. Producer B. Primary consumer C. Secondary consumer D. Decomposer	A. Producer	Producers occupy the first trophic level because they make their own food, usually through photosynthesis, and provide energy to organisms at higher trophic levels.
Trophic levels	Identify which of the things found in the environment are biotic or abiotic	Remembering	6	Rose Marianne Louis M. Gardoce	_____ 6. Which organism is usually found at the second trophic level ? A. Producer B. Primary consumer C. Secondary consumer D. Tertiary consumer	B. Primary consumer	Primary consumers occupy the second trophic level because they obtain energy by eating producers.

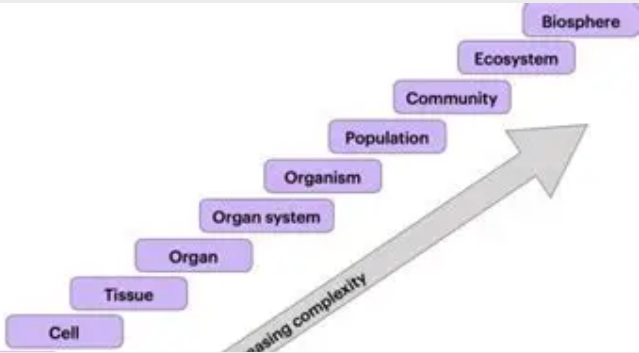
Ecological Relationships – Symbiotic	Identify which of the things found in the environment are biotic or abiotic	Remembering	7	Rose Marianne Louis M. Gardoce	_____ 7. What type of symbiotic relationship benefits both organisms? A. Parasitism B. Commensalism C. Mutualism D. Predation	C. Mutualism	In mutualism, both organisms benefit from their interaction.
Ecological Relationships – Symbiotic	Identify which of the things found in the environment are biotic or abiotic	Remembering	8	Rose Marianne Louis M. Gardoce	_____ 8. What type of relationship occurs when one organism benefits while the other is neither helped nor harmed? A. Mutualism B. Commensalism C. Parasitism D. Competition	B. Commensalism	In commensalism, one organism benefits while the other is neither helped nor harmed.
Ecological Relationships – Symbiotic	Identify which of the things found in the environment are biotic or abiotic	Remembering	9	Cherrie Anne B. Masculino	_____ 9. Which symbiotic relationship benefits one organism while harming the other? A. Mutualism B. Commensalism C. Parasitism D. Competition	C. Parasitism	In parasitism, one organism benefits while the other organism is harmed.
Ecological Relationships – Symbiotic	Identify which of the things found in the environment are biotic or abiotic	Remembering	10	Cherrie Anne B. Masculino	_____ 10. Which situation is an example of parasitism? A. A bee obtains nectar from a flower while pollinating it.	C. A tick feeds on the blood of a dog.	The tick benefits by obtaining nourishment, while the dog is harmed by the parasite.

					B. A remora attaches to a shark and feeds on leftover food. C. A tick feeds on the blood of a dog. D. Two plants compete for sunlight.		
Ecological Relationships – Non Symbiotic	Describe different ecological relationships found in the ecosystem	Remembering	11	Cherrie Anne B. Masculino	_____ 11. Which ecological relationship occurs when two organisms compete for the same limited resource? A. Predation B. Competition C. Mutualism D. Parasitism	B. Competition	Competition occurs when organisms seek the same limited resource, such as food, space, water, or sunlight.
Ecological Relationships – Non Symbiotic	Describe different ecological relationships found in the ecosystem	Remembering	12	Cherrie Anne B. Masculino	_____ 12. Which type of ecological relationship occurs when one organism captures and consumes another organism? A. Competition B. Commensalism C. Predation D. Mutualism	C. Predation	Predation occurs when one organism, the predator, captures and eats another organism, the prey.
Ecological Relationships – Non Symbiotic	Describe different ecological relationships found in the ecosystem	Remembering	13	Cherrie Anne B. Masculino	_____ 13. Which situation is an example of competition? A. Two plants growing in the same area compete for sunlight. B. A spider catches and eats a fly.	A. Two plants growing in the same area compete for sunlight.	The plants are competing for a limited resource, which is sunlight.

					C. A tapeworm lives inside a human intestine. D. A bee obtains nectar from a flower.		
Ecological Relationships – Non Symbiotic	Describe different ecological relationships found in the ecosystem	Remembering	14	Cherrie Anne B. Masculino	_____ 14. In the relationship between a lion and a zebra, which organism is the prey? A. The lion B. The zebra C. Both organisms D. Neither organism	B. The zebra	The zebra is the organism hunted and consumed by the lion, making it the prey.
Transfer of energy (food chain, food web, energy pyramid)	Describe how energy is transformed through feeding relationships	Remembering	15	Cherrie Anne B. Masculino	_____ 15. In a food chain, which organisms obtain energy by making their own food? A. Consumers B. Producers C. Predators D. Decomposers	B. Producers	Producers make their own food and serve as the starting point of energy transfer in a food chain.
Transfer of energy (food chain, food web, energy pyramid)	Describe how energy is transformed through feeding relationships	Remembering	16	Cherrie Anne B. Masculino	_____ 16. Which organisms obtain energy directly by feeding on producers? A. Primary consumers B. Secondary consumers C. Decomposers D. Tertiary consumers	A. Primary consumers	Primary consumers obtain energy directly from producers by feeding on them.

Transfer of energy (food chain, food web, energy pyramid)	Describe how energy is transformed through feeding relationships	Remembering	17	Prince Nimrod C. Cantillo	_____ 17. What does the arrow pointing from one organism to another in a food chain represent? A. Transfer of carbon dioxide B. The flow of energy C. The physical movement of the predator D. The direction of water flow	B. The flow of energy	In a food chain or food web, arrows are explicitly used to indicate the direction in which energy flows from the organism being consumed to the consumer.
Transfer of energy (food chain, food web, energy pyramid)	Describe how energy is transformed through feeding relationships	Remembering	18	Prince Nimrod C. Cantillo	_____ 18. Which of the following provides the initial source of energy for almost all food webs on Earth? A. Soil nutrients B. Water C. The Sun D. Decomposers	C. The Sun	The sun is the ultimate source of energy for most ecosystems, as primary producers require solar energy to synthesize food.
Transfer of energy (food chain, food web, energy pyramid)	Describe how energy is transformed through feeding relationships	Remembering	19	Prince Nimrod C. Cantillo	_____ 19. What is the scientific term used to describe each step or level in a food chain or energy pyramid? A. Ecosystem tier B. Trophic level C. Biomass step D. Consumer bracket	B. Trophic Level	A trophic level is defined as the specific position or step an organism occupies within a food chain, food web, or energy pyramid.
Transfer of energy (food chain, food web, energy pyramid)	Describe how energy is transformed through feeding relationships	Remembering	20	Prince Nimrod C. Cantillo	_____ 20. According to the standard rule of energy transfer in an energy pyramid, approximately what percentage of energy is passed on to the next level? A. 10% B. 50% C. 90% D. 100%	A. 10%	The 10% rule is a foundational ecological concept stating that only about 10% of the energy from one trophic level is successfully transferred to the subsequent level.

Levels of biological organization	Use a labeled diagram to describe the connections between the levels of biological organization to one another from cells to the biosphere	Applying	21	Prince Nimrod C. Cantillo	_____ 21. Examine the flowchart below: Cell -> X -> Organ -> Organ System -> Organism Which of the following best represents “X”? A. Biosphere B. Organelle C. Population D. Tissue	D. Tissue	A tissue is a group of similar cells working together to perform a specific function, bridging the structural level between a single cell and a complete organ.
Levels of biological organization	Use a labeled diagram to describe the connections between the levels of biological organization to one another from cells to the biosphere	Applying	22	Prince Nimrod C. Cantillo	_____ 22. A student is tasked to draw a diagram showing how a localized group of organisms connects to the broader environment. If the diagram starts with “Population,” what is the next broader level they must draw immediately after? A. Community B. Organism C. Biosphere D. Biome	A. Community	In the hierarchy of biological organization, a community (multiple interacting populations in a specific area) is the next broader level immediately following a single population.
Levels of biological organization	Use a labeled diagram to describe the connections between the levels of biological organization to one another from cells to the biosphere	Applying	23	Prince Nimrod C. Cantillo	_____ 23. While constructing a graphic organizer of the human body, a student groups the stomach, intestines, and liver under one main heading. Which level of biological organization should be used as the title for this main heading? A. Cell B. Tissue C. Organ System	C. Organ System	The student is applying the concept of biological hierarchies by grouping individual organs (stomach, intestines, liver) under their collective functional category, which is an organ system.

					D. Organism		
Levels of biological organization	Use a labeled diagram to describe the connections between the levels of biological organization to one another from cells to the biosphere	Applying	24	Prince Nimrod C. Cantillo	_____ 24. In a science project, a student is asked to draw a layered diagram of a marine environment. The student draws fish and corals in one layer, and adds water currents, temperature, and sunlight in the final layer. What specific level of biological organization does this final layer represent? A. Population B. Community C. Ecosystem D. Biosphere	C. Ecosystem	By including nonliving environmental factors (water currents, temperature, sunlight) alongside the living community (fish, corals) in their diagram, the student is representing an ecosystem.
Levels of biological organization	Use a labeled diagram to describe the connections between the levels of biological organization to one another from cells to the biosphere	Applying	25	Arnold Daniel B. Jaramillo	For numbers 25-26. View the attached diagram.  _____ 25. A pack of wolves are a group of wolves that live and hunt together. However sometimes, wolves can travel alone in order to find mates. What level of organization do lone wolves belong to ? A. Organism B. Community C. Ecosystem D. Biosphere	A. Population	By analyzing the given diagram and its different levels, the learner can figure out which level of organization the lone wolf belongs to and since it is a single wolf, it is an organism.

Levels of biological organization	Use a labeled diagram to describe the connections between the levels of biological organization to one another from cells to the biosphere	Applying	26	Arnold Daniel B. Jaramillo	_____ 26. The levels of organization get more complex and involve more organic life the higher it goes. Based on that, what level of organization would the entire planet belong to? A. Population B. Community C. Ecosystem D. Biosphere	D. Biosphere	By analyzing the given diagram and the given statement. The learner should conclude that since the higher the organization level goes the more organisms are involved in it and that the highest level is a biosphere. Then the planet that involves ALL organisms should be a biosphere
Effects of human activity in the ecosystem	Predicts effects of changes in the ecosystems	Understanding	27	Arnold Daniel B. Jaramillo	_____ 27. What would happen to an ecosystem if it lost its producers? A. The ecosystem would remain the same B. The ecosystem would die due to lack of food C. The ecosystem would thrive with more animals D. The ecosystem would thrive with a lot of plants	B. The ecosystem would die due to lack of food	Producers or plants are the main source of energy of the ecosystem due to their ability to collect the energy of the sun. Having no producers means there would also be no energy in the form of food to be distributed in the ecosystem, hence it would slowly die off.
Effects of human activity in the ecosystem	Predicts effects of changes in the ecosystems	Understanding	28	Arnold Daniel B. Jaramillo	_____ 28. How does illegal mining affect an ecosystem even though it doesn't kill animals? A. It doesn't affect the ecosystems B. It affects the abiotic factors that ecosystems need to survive C. It affects the ecosystems by shaking the soil underneath during mining D. It affects ecosystems by making it taking the minerals away from the soil	B. It affects abiotic factors that ecosystems need to survive	Soil and water are just a few abiotic factors that ecosystems need in order to survive. Mining affects both of these factors therefore indirectly affecting the ecosystems as well.

Ecosystem conservation	Explain the importance of collective actions and preserving ecosystems	Understanding	29	Arnold Daniel B. Jaramillo	_____ 29.How would recycling plastics help a marine ecosystem? A. It wouldn't help a marine ecosystem B. The marine ecosystem wouldn't thrive because there is already trash in the ocean C. The marine ecosystem would thrive because marine life can use the recycled trash D. The marine ecosystem's abiotic and biotic factors would not be interrupted by trash	D. The marine ecosystem's abiotic and biotic factors would not be interrupted by trash	If majority of the people would recycle there would be less trash in the ocean, less trash would help organisms and abiotic factors in the ecosystem perform their purpose without being disturbed or negatively affected by trash
Ecosystem conservation	Explain the importance of collective actions and preserving ecosystems	Understanding	30	Arnold Daniel B. Jaramillo	_____ 30. Is it environmentally better for people to ride public transportation than to drive a car solo everyday? If yes, explain why? A. Yes. Public transportation can allow more people to travel while producing less green house gas B. Yes. Public transportation allows people to spend less money and energy compared to buying their own cars C. No. Public transportation isn't better because there's still going to be cars the people ride when they go to work. D. No. Public transportation isn't better because it is harder for people to travel by commute and it is easier for them to have their own vehicles	A. Yes. Public transportation can allow more people to travel while producing less green house gas	Even if we can't fully stop using cars, reducing the amount of cars by using public transportation can greatly benefit our environment.

Use the template below.



Rubric

CRITERIA & WEIGHT PERCENTAGE	EXCELLENT	VERY SATISFACTORY	SATISFACTORY	NEEDS IMPROVEMENT	RATING
	4	3	2	1	
1. Promptness in the submission of an objective test (10%)	Submitted on or before the deadline	Submitted 1 day after the deadline	Submitted 2 days after the deadline	Submitted 3 or more days after the deadline	
2. Completeness (20%) a. Heading (Title/Quarterly Test in _____) b. Other details (date submitted, group name, list of members. etc.) c. 30 items written d. Items written used in multiple-choice format e. Key to corrections f. Draft (template) and the final form of the test g. Others	Satisfies ALL of the given criteria	4-5 of the given criteria are satisfied	2-3 of the given criteria are satisfied	1 or none of the given criteria is satisfied	
3. Quality of the final test (30%) 3.1 Over-All a. All test parts have well-stated and complete directions b. Items are aligned with the Table of Specifications c. Desired format, alignment, and spacing, etc. are attained (Face Validity) d. Content/ Concepts are accurate (Content Validity) e. Others	Satisfies ALL of the given criteria	3 of the given criteria are satisfied	2 of the given criteria are satisfied	1 or none of the given criteria is satisfied	

3.2 Guidelines in Test Construction (30%) a. All items are well-written (i.e. free from any violation of the guidelines in test construction)	1 guideline in test construction was not followed	2-3 guidelines in test construction were not followed	4-5 guidelines in test construction were not followed	6 or more guidelines in test construction were not followed	
4. Others (10%) 1. Computerized 2. Neat 3. Error-free (grammar, spelling, and erasures) 4. Others	Satisfies ALL of the given criteria	2 of the given criteria are satisfied	1 of the given criteria is satisfied	None of the given criteria is satisfied.	

Interpretation: 3-4= 90-100% 2-2.9= 80-89% 1-1.9= 70-79%

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Adapted from “Ana-Holistic Scoring Rubric for Portfolios”, by Dr. M.U.Balagtas (2011)